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Hydrolysis of bamboo biomass by subcritical water treatment

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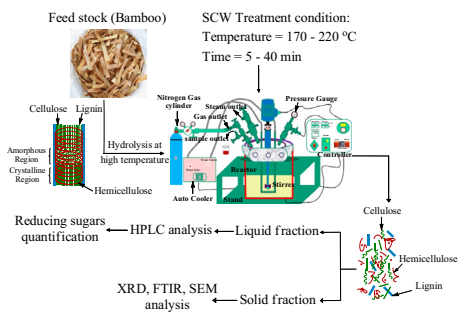
HIGHLIGHTS

- Reaction temperature and time shown more impact on total reducing sugars yield.
- Highest yield of total reducing sugars obtained at 180 °C and 25 min of reaction time.
- X-ray diffraction profile shown increase in crystallinity index after treatment.
- Kinetics of bamboo was considered as first-order irreversible series reaction.

GRAPHICAL ABSTRACT



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ABSTRACT

The aim of present study was to obtain total reducing sugars (TRS) from bamboo under subcritical water (SCW) treatment in a batch reactor at the temperature ranging from 170 °C to 220 °C and 40 min hydrolysis time. Experiments were performed to investigate the effects of temperature and time on TRS yield. The maximum TRS yield (42.21%) was obtained at lower temperature (180 °C), however longer reaction time (25 min). X-ray diffraction (XRD), Fourier transform infrared (FTIR) spectroscopy, and scanning electron microscopy (SEM) analysis were used to characterise treated and untreated bamboo samples. The XRD profile revealed that crystallinity of bamboo increased to 71.90% with increase in temperature up to 210 °C and decreased thereafter to 70.92%. The first-order reaction kinetic model was used to fit the experimental data to obtain rate constants. From the Arrhenius plot, activation energy and pre-exponential factor at 25 min time were found to be 17.97 kJ mol^{-1} and 0.154 min^{-1}, respectively.

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1. Introduction

The continuous increase in the population growth pushes towards expanding the utilisation of fossil assets in the production of energy and chemicals (Mohan et al., 2015b). The high utilisation rate of fossil resources has a direct impact on the economic, environmental, political, life quality of population and also increases emission of greenhouse gases, which aggravate global warming. Hence it requires an alternative energy solution to decrease today's rapid consumption of fossil resources. A potential alternative solution to this problem could be utilisation of lignocellulosic biomass as an alternative energy source for the future (Holm and Lassi, 2011). Currently extensive research is being undertaken to produce environmentally friendly energy power and worth included chemicals from lignocellulosic biomass (Mohan et al., 2015a, 2015b). Use of biomass for energy avoids the increase of CO_2 in the atmosphere (Asghari and Yoshida, 2010). As per the facts, aggregate energy prepared from photosynthesis is ten times higher than that of the fossil-fuel assets utilised within the world consistently (Wang et al., 2010).

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Lignocellulosic biomass is dominated by three naturally occurring polymers, cellulose (30–50%), hemicellulose (25–35%) and lignin (5–30%). The sources of lignocellulosic biomass, which are waste agriculture residue, wood, grass; forestry and municipal solid wastes are recognised as tempting feedstock for the creation of fuel alcohol, due to their accessibility in huge amounts at low price (Roque et al., 2012). Cellulose is the primary constituent in any biomass to produce oligosaccharides (cellotetraose, cellotriose, and cellobiose), and monomeric sugars (glucose and fructose). Hemicellulose is the secondary most abundant polymer in biomass to produce pentose sugars (xylose, arabinose) and hexoses (galactose, glucose and mannose). Hemicellulose sugars have a lower molecular weight compared to cellulose and limb with short horizontal chains that are effortlessly hydrolysed (Agbor et al., 2011). The primary point of lignocellulosic biomass pretreatment is to alter the cellulose structure, and make the cellulose structure more accessible to enzymes and/or chemicals (Holm and Lassi, 2011). Among the various biomass wastes, bamboo is an ancient woody grass and dispersed in tropical, subtropical and mild temperature zones. Generally, bamboo seen as the "poor man's tree," in recent years it has climbed to an innovative, modern crude material and substitute for wood. In Asia, India is the second richest country in bamboo genetic assets after China. These two nations together have more than half portion of the aggregate bamboo resources universally (Katwal et al., 2003). For the management and maintenance of natural resources, 170 states have signed on the action program for the 21st century (Agenda 21) in the year 1992 in Brazil. The aim is to search for new solutions to decrease today's rapid consumption of fossil resources and non-renewable resources such as petroleum, natural gas, coal and minerals (Kamm and Kamm, 2004).

The significant innovations proposed for hydrolysis of lignocellulosic biomass incorporate acid or base treatments, which are not satisfactory because of the ecological issues, and enzymatic hydrolysis, which is a time taking process (Rogalinski et al., 2008). Whereas, sub or supercritical water treatment have been picking up an expanding consideration as an environmentally friendly technique with an extensive variety of provisions, such as extraction, hydrolysis, and wet oxidation of organic compounds. It is very cheap, non-toxic, non-flammable, and non-explosive and have many advantages compared to other substances, mainly in the field of "green chemistry" (Ehara and Saka, 2005; Zhao et al., 2009). Sub critical water (SCW) is defined as hot water at temperature ranging between 100 and 374 °C under high pressure to maintain water in the liquid state. The physical properties of water in sub critical condition changes with increase in temperature (Aida et al., 2010; Tsigie et al., 2012). SCW treatment has been known as an exceptionally extraordinary medium or solvent for various chemical reactions and extraction of compounds. SCW is also used as an effective catalyst for hydrolysis or biodegradable reactions. A wide variety of organic pollutants from industrial waste is completely removed by SCW within a few minutes. Therefore, SCW is found to be turning into a promising extractant and catalyst (Pourali et al., 2009).

Scanty information is available in the literature on SCW treatment of bamboo biomass. In this present study, SCW treatment has been employed as a method to produce total reducing sugars (TRS) from bamboo at temperature ranging from 170 °C to 220 °C and reaction time 5 to 40 min. Treated and untreated bamboo samples are characterised by X-ray diffraction (XRD), Fourier transform infrared (FTIR) spectroscopy, and scanning electron microscopy (SEM) techniques to analyse the structural changes. Attempts have been made to investigate the reaction kinetics of bamboo to fit the experimental data for obtaining the reaction kinetic parameters (i.e., rate constant, activation energy and pre-exponential factor). The study also helps in understanding the potential usage of bamboo as a feedstock for TRS production.

2. Methods

2.1. Materials

Bamboo biomass (*Bambusa cacharensis* R.B. Majumdar) used as a raw material in this study was obtained from nearby IIT Guwahati forest area, Assam, India (26°11′14″N 91°41′30″E). The air-dried biomass samples were chopped using chopping machine, and sieved (mesh size – BSS 30) to get homogeneous powder (particles size 0.5 mm). Before the experiments, bamboo powder was dried at 60 °C for 24 h. The high performance liquid chromatography (HPLC) standards, glucose (⩾99.5%), xylose (⩾99%), fructose (⩾99%), cellobiose (⩾98%), arabinose (⩾98%), 5-(hydroxyme thyl)furfural (5-HMF, ⩾99%), furfural (⩾98%) and glycol (⩾99%) were purchased from Sigma–Aldrich.

2.2. Experimental procedure

The hydrolysis of bamboo biomass was carried out in a high pressure reactor (maximum working pressure 350 kg cm^{-2} and temperature 500 °C, Model 1734, Amar Equipment Pvt. Ltd., India) equipped with a stainless steel 500 ml reaction vessel (160 mm height, 75 mm internal diameter). The reactor is equipped with a magnetic driven stirrer with 60 ml minimum stirring volume, temperature controller, and U-type internal cooling coil. The reactor was heated with the help of 2.0 kW electric heaters. Six HEX M8 × 70 screws which can afford 12.8 tons of tensile force were used for tightening the reactor with its cap. The hydrolysis process was run under batch mode. For SCW treatment, nitrogen gas (98% purity, purchased in Guwahati, Assam, India) was used to maintain the constant pressure inside the reactor. The bamboo hydrolysis process was performed between 170 °C and 220 °C (interval of 10 °C) for 5–40 min reaction time (time interval 5 min), at a pressure corresponding to the vapour pressure curve at a given temperature. The reaction content was mixed continuously via magnetic driven stirrer at desired temperature (Ozturk et al., 2010).

Experiments were performed in a 500 ml reaction vessel using 5 g of bamboo in 100 ml of deionised water and were kept constant unless otherwise stated. At the beginning to maintain the constant hydrolysate volume, an additional 20 ml of deionised water was added to the system considering the water loss due to evaporation (~10%) and entrainment in the biomass residue (~5%). Temperature inside the reactor was measured using a thermocouple and controlled at operating temperature. The pressure inside the reactor was maintained at 7.9, 10, 12.5, 15.5, 19 and 23.1 bar and the corresponding recorded temperatures were 170, 180, 190, 200, 210, and 220 °C, respectively, or slightly above by using nitrogen as a pressurised gas to ensure that water remained as liquid (Pinkowska et al., 2011; Tsigie et al., 2012). At these conditions, liquid samples were collected through sample outlet at different time intervals, and immediately quench to provide truly representative samples.

In case of biomass residue, samples were cooled up to room temperature within the reactor. Finally, treated biomass was collected and dried at 80 °C for 24 h. The treated dried samples were characterised by XRD, FTIR and SEM analysis. The collected liquid samples were analysed by HPLC. All the experiments were performed in duplicate, and the average values are reported. After

each experiment, the reactor was flushed with water (~pressure 15 bar) to avoid chocking of sample port and gas inlet port by residual biomass.

2.3. High performance liquid chromatography (HPLC) analysis

After collecting the liquid samples at different time interval, samples were centrifuged, and filtered with 0.2 μm nylon membrane filter paper (CAT NO: 170025R, Diameter: 25 mm purchased from AXIVA, India) through 0.22 μm syringe filter before HPLC analysis. TRS and degraded products were analysed by HPLC (Perkin Elmer Series 200, USA). The HPLC equipment consists of a pump, an auto sampler, vacuum degasser, network chromatography interface (NCI) 900, and connected with both refractive index (RI) and UV/Vis detectors. The column used for HPLC analysis was SUPELCOGEL Ca with dimensions 300 mm of length (*L*), 7.8 mm of inner diameter (I.D) (Cat No: 59305-U), and connected to SUPELCOGEL Ca* guard column (*L* and I.D: 50 mm × 4.6 mm; Cat. No: 59306-U), which were purchased from Sigma–Aldrich. HPLC instrument was operated at an oven temperature of 80 °C at 0.8 ml min^{-1} flow rates for total run time of 40 min, and milli-Q water used as a mobile phase. RI detector was used to provide quantitative analysis of the products obtained in hydrolysis process. The HPLC peaks were recognised by the comparison of sample peak retention time with the help of calibration plots for pure compounds (Kabyemela et al., 1999; Ju et al., 2011; Cruz et al., 2012).

2.4. Fourier transforms infrared spectroscopy (FTIR)

FTIR spectroscopy (Shimadzu, IR Affinity 1, Japan) was used to study the chemical changes in the treated and untreated bamboo samples. 20 mg of bamboo sample was mixed with 30 mg of spectroscopic grade dried KBr. After mixing sample was analysed using an FTIR spectroscopy over the wave number range of 4000–400 cm^{-1}, with a resolution of 2 cm^{-1} and 30 scans per sample. The background spectrum of pure dried KBr was subtracted from the sample spectrum (Ozturk et al., 2010; Ninomiya et al., 2013). Baseline and other necessary corrections for penetration depth and frequency variations are applied using Shimadzu IR solution 1.5 software.

2.5. X-ray diffraction (XRD) analysis

The crystallinity of treated and untreated bamboo (i.e., crystallinity of cellulose in bamboo) was measured by XRD technique (Brucker D8 Advance X-ray diffraction measurement system, Japan). Rigaku miniflex powder XRD was equipped with Cu Kα (λ = 1.541 Å) radiation settled at 40 kW voltages, 40 mA current. The range of diffraction angle (2θ) is in between 10 and 30° at a scan speed of 1° min^{-1}, and step size 0.05°. The crystallinity index (CrI) of cellulose in bamboo was calculated based on diffraction intensities of crystalline and amorphous regions of cellulose using following Eq. (1) (Kumar et al., 2010; Ozturk et al., 2010; Ninomiya et al., 2013):

$$\mathrm{CrI} = \frac{I_{002} - I_{\mathrm{amr}}}{I_{002}} \times 100 \quad (1)$$

where I_{002} = peak intensity of crystallinity region of 002 lattice plane at 2θ = 22.4°, and I_{amr} = peak intensity of amorphous region at 2θ = 18.5°.

2.6. Scanning electron microscopy (SEM)

Surface morphology of bamboo samples, before and after hydrolysis was qualitatively studied using SEM (LEO, Model 1430vp, Germany). The dry samples were fixed with a conducting adhesive carbon tape on an aluminium stub followed by 2–3 times sputter coating of gold. Finally the gold coated samples were observed in SEM, which was operating at 15 kV (Kumar et al., 2010; Ozturk et al., 2010; Roque et al., 2012).

3. Results and discussion

The major components of bamboo (i.e., cellulose and hemicellulose) hydrolyse under SCW condition to produce TRS (i.e., glucose, cellobiose, fructose, xylose and arabinose etc.) at higher temperatures in the range of 170–220 °C, and 5–40 min of reaction time. The TRS obtained during the process can be converted to alcohol via fermentation using micro-organisms or yeast. Reaction time and temperature are the major parameters, which affect the yield of TRS from bamboo under SCW treatment. In this study, the major focus was on to maximise the TRS yield from bamboo, which contains 38.3% cellulose and 25.51% hemicellulose. The yield of reducing sugar is calculated by measuring the sugar concentration in hydrolysate multiply by obtained hydrolysate volume. The total mass of sugar obtained was then divided by initial mass of bamboo treated. However, during the hydrolysis process sample volume of 1.5 ml was withdrawn at each reaction time interval. Therefore the volume of water remained during the hydrolysis was considered as 103, 101.5, 100, 98.5, 97, 95.5, 94, and 92.5 ml at the corresponding time period 5, 10, 15, 20, 25, 30, 35, and 40 min.

3.1. Effect of reaction temperature on total reducing sugar yield

The hydrolysis of bamboo under SCW treatment was studied at the temperatures ranging from 170 °C to 220 °C, and the effect of reaction temperatures on TRS yield is shown in Fig. 1. From the Fig. 1, it can be seen that yield of TRS tends to increase with increasing temperature from 170 °C to 180 °C, further increase in the temperature resulted in decreasing TRS yield. The maximum yield of TRS was found to be 42.21% at 180 °C and 25 min of reaction time. From the individual sugars plot (Fig. 2), it was observed that at 180 °C, the yield of hemicellulose sugars (i.e., xylose and arabinose) was higher than the cellulose sugars (i.e., cellobiose, glucose and fructose), which might be due to faster decomposition rate of hemicellulose compared to cellulose. These results may be attributed to the presence of more crystalline nature of cellulose structure. As a result, for obtaining higher yield of glucose, and

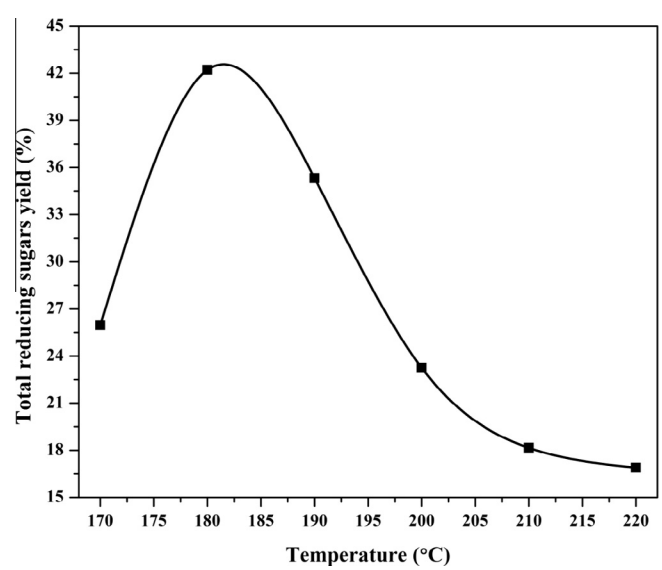

Fig. 1. Effect of reaction temperatures on the total reducing sugars yield at 25 min.

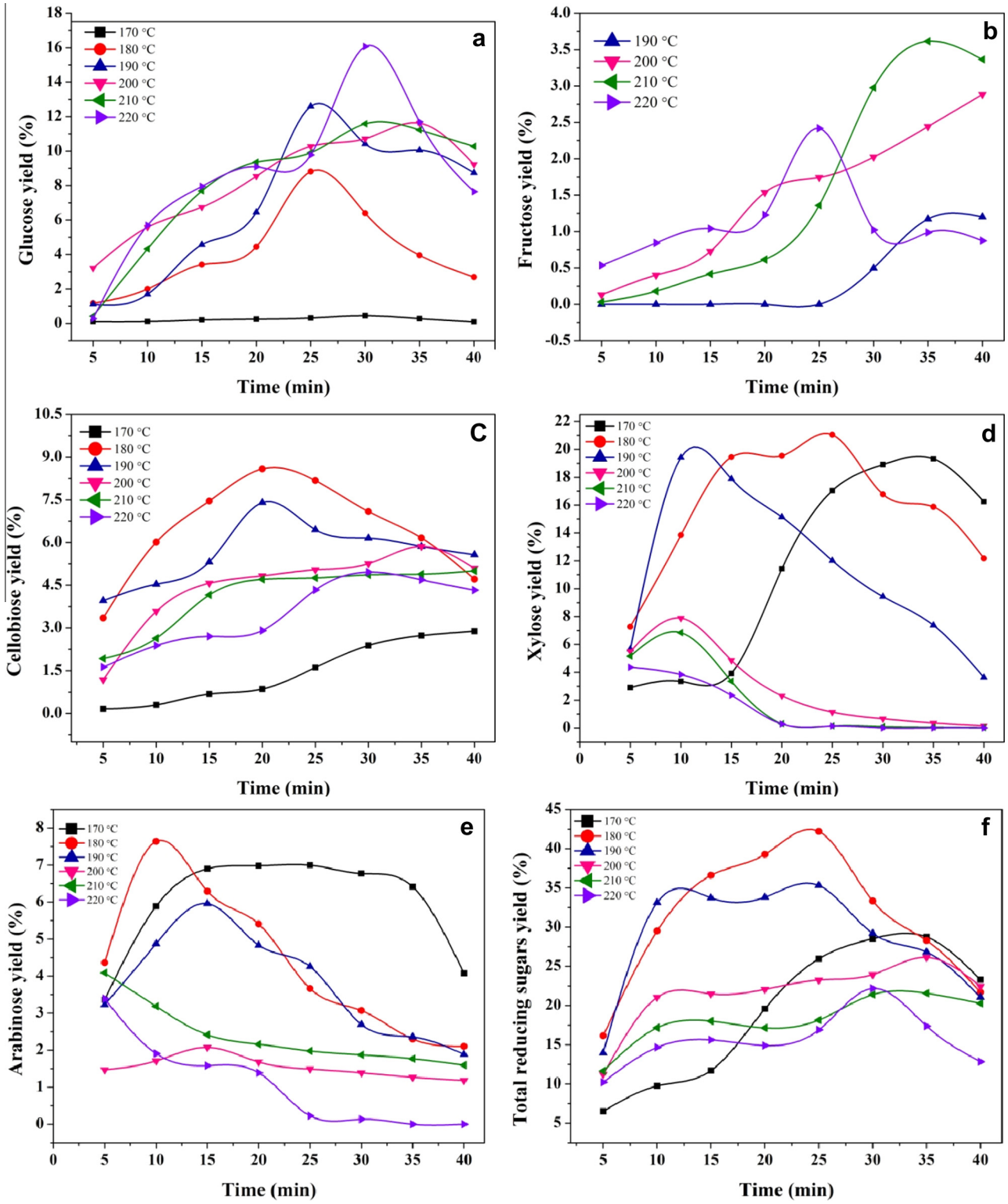

Fig. 2. Yield of sugars obtained from bamboo as a function of reaction temperatures and time: (a) glucose, (b) fructose, (c) cellobiose, (d) xylose, (e) arabinose, and (f) total reducing sugars.

fructose, cellulose needs higher temperature to break down into monomeric sugars. At higher temperatures, the decomposition rates of glucose or oligomers (cellotetraose, cellotriose, and cellobiose) are higher than the cellulose hydrolysis rate (Zhao et al., 2011). This may be also one of the reasons for not attaining high yield of monomeric (glucose, and fructose) sugars. With the increase in temperature, the degree of polymerisation of the solubilised oligosaccharides decreases. This may be due to increase in ionisation constant of water (K_w) with increasing temperature (Mochidzuki et al., 2003; Tanaka et al., 2012). The increase in concentration of ions, not only helps in destroying the glycosidic bonds in the cellulose, but also provides very short time for the

conversion of oligosaccharides to monomers (Rubio et al., 1998). Lu and Saka (2010) have studied the hydrolysis of Japanese beech (*Fagus crenata*) under batch and semi-flow water treatment at subcritical temperature and pressure. It was observed that, the maximum sugars were obtained at 190 °C in a batch treatment compared to semi-flow water treatment, although the semi-flow treatment, reaction temperature was 250 °C. In the present study, maximum TRS yield was obtained at 180 °C which is comparatively lower temperature than the reported temperature value.

Fig. 2(a–f) shows the yield of individual sugars and TRS at different temperatures. Fig. 2a shows that increase in the temperature from 170 °C to 220 °C facilitated the yield of glucose, and also attained maximum (16.07%) at 220 °C and 30 min reaction time. Moreover, glucose can also isomerise to fructose or may get degraded to 5-HMF, and other products (Kabyemela et al., 1999). On the other hand, fructose is less stable than glucose at SCW conditions (Pourali et al., 2009). In the hydrolysis process, yield of fructose increases with increasing temperature from 190 °C to 210 °C i.e., 3.61% at 210 °C and 35 min reaction time (Fig. 2b), and subsequently decreased. In the same way, cellobiose yield increased as the temperature of the system increased from 170 °C to 180 °C (9.28% at 180 °C and 20 min), and from then on decreased with increase in the process temperature (Fig. 2c). It implies that, cellobiose decomposed (C_6–C_6 sugar) to monomeric sugar (C_6 sugar). From Fig. 2c and a, it was noticed that the yield of cellobiose is more than the glucose yield up to 180 °C, and thereafter cellobiose yield decreased with increase in the temperature. This resulted in the enhancement of glucose and other product formations due to the decomposition of cellobiose at higher temperatures. The highest yield of hemicellulose sugars such as xylose (21.05%) and arabinose (7.64%) was achieved at 180 °C (Fig. 2d and e). Further increase in the temperature, decomposition of xylose and arabinose is more in the bamboo hydrolysis process. In the hydrolysis process at 180 °C, compared to other sugars, the yield of xylose (major component in the hemicellulose) attained was higher, which might be due to the faster decomposition of hemicellulose at lower temperature under SCW conditions. Even the formation and degradation rates of the reducing sugars vary with the type of monomers. The formation of degraded products such as 5-HMF, furfural, glycol etc. was found to be increased with increasing reaction time and temperature (Supplementary Figs. S.1–S.3).

3.2. Effect of reaction time on reducing sugar yield

The effect of reaction time on TRS yield was investigated under SCW condition (Fig. 2f). During the experimental run lasting for 40 min of reaction time, after every 5 min sampling was done. Hydrolysis process beyond 40 min was not studied to avoid the decomposition of sugars. Fig. 2f shows that yield of TRS increased with an increase in the reaction time up to certain period. However, further increase in the reaction time decreases TRS yield. The decrease in TRS yield may be due to further decomposition of reducing sugars to other degraded products such as 5-HMF, furfural, glycol, etc. (Aida et al., 2010; Prado et al., 2014). From the obtained results, it can be seen that at 180 °C and 25 min reaction time, highest yield of TRS (42.21%) was achieved (Fig. 2f). Further increase in the reaction time beyond 25 min decreases TRS yield. This could be due to the formation of degraded products. However, glucose yield obtained at 180 °C and 220 °C at different hydrolysis time i.e., 25 and 30 min was found to be 8.81% and 16.07% and for xylose 21.05% and 0.004%, respectively. Further increase in the reaction time resulted in the conversion of degraded products (5-HMF, furfural, glycol etc). The simplified reaction pathway of lignocellulosic biomass hydrolysis under SCW condition is shown in Fig. 3.

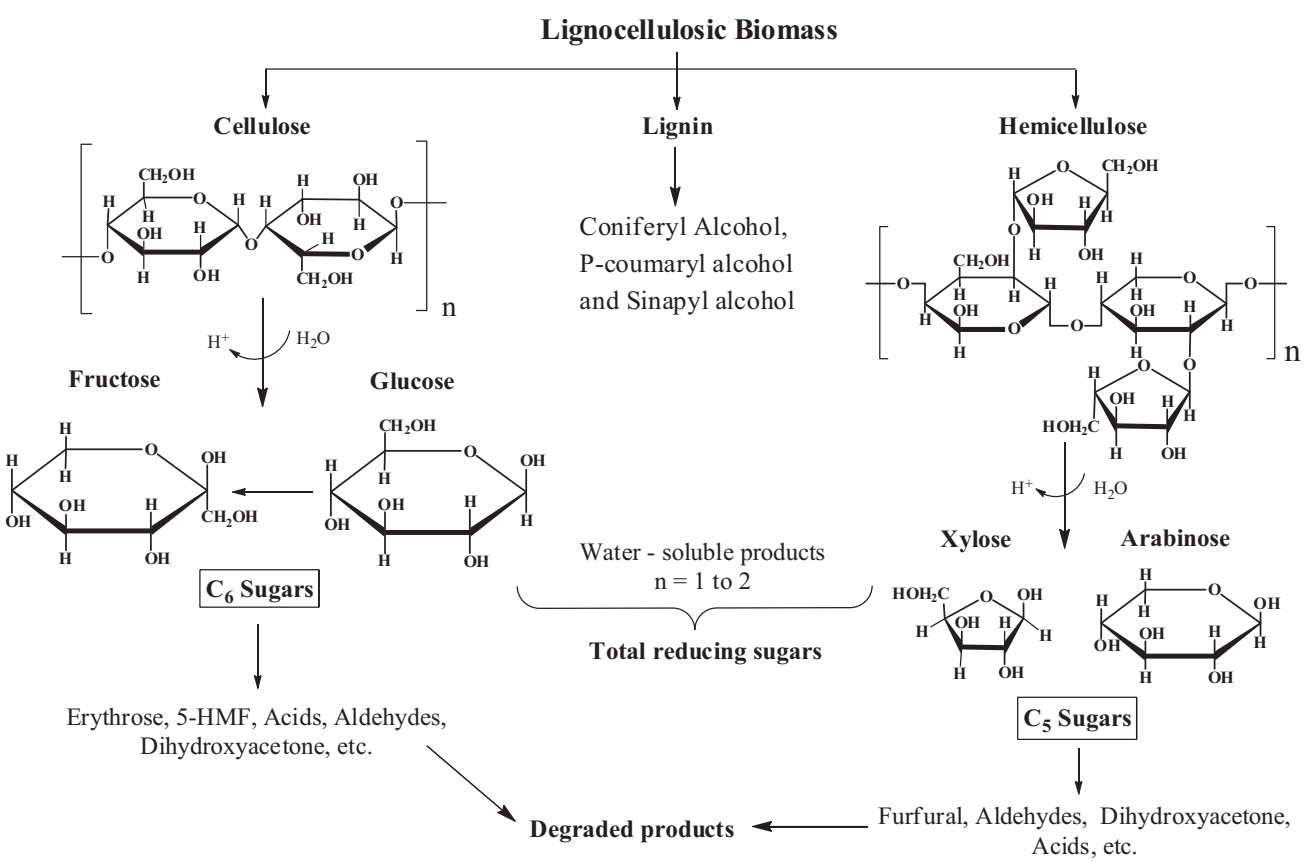

Fig. 3. Simplified reaction pathway of lignocellulosic biomass hydrolysis under subcritical water treatment (adapted from Aida et al. (2010) and Kumar et al. (2010)).

Figs. 1 and 2 reveals that highest yield of TRS was obtained at lower temperature (180 °C), however longer reaction time (25 min). For this reason, reaction kinetics was carried out at two different reaction times, (1) total reaction time (i.e., 40 min) and (2) the reaction time at which highest yield was obtained (i.e., 25 min at 180 °C, see Fig. 2f). This reaction kinetics is important for better understanding the hydrolysis of biomass under SCW conditions.

3.3. Comparison of cellulose in bamboo and pure cellulose decomposition to total reducing sugars

Fig. 4 shows the comparative plot of TRS yield obtained from cellulose fraction in bamboo and pure cellulose as a function of temperature. Most of the products formed during this process are hydrolysis products such as glucose, fructose, and cellobiose accounted altogether as sugars of cellulose. From Fig. 4, it can be observed that at a fixed hydrolysis time i.e., 20 min, the TRS yield of pure cellulose increased with an increase in the temperature from 150 °C to 200 °C. Further increase in the temperature beyond 200 °C (i.e., 225 °C and 250 °C) resulted in decreased TRS yield due to higher degradation rate of formed sugars. To give a quantitative idea for temperature 200 °C and 220 °C , the maximum TRS yield obtained for pure cellulose was 45.04% at 200 °C and 20 min of hydrolysis time, whereas for cellulose fraction in bamboo it was 22.05% at slightly higher temperature and time i.e., 220 °C and 30 min. However, yield of TRS from cellulose fraction in bamboo first increased and reached a maximum value (22.05% at 220 °C) up to 30 min and then started decreasing with time (Fig. 4). On comparing the TRS yield for cellulose fraction in bamboo and pure cellulose at 200 °C and 20 min, highest TRS yield of 14.91% and 45.04% was achieved, further increase in the temperature decreased the TRS yield in both the samples (Fig. 4). Similar results with marginal variation in the extent of TRS yield were also obtained for pure cellulose. It has been observed that the production of hexose sugars mainly influenced by the reaction time and temperature. Due to hard structure of bamboo biomass, the hydrolysis of cellulose fraction in bamboo occurred at slightly higher temperature and time compared to pure cellulose. Comparison of the obtained results with literature reveals that, no report is available in the literature on hydrolysis of bamboo under SCW treatment hence the results of the present study have been compared with the biomass having almost similar composition. Prado et al. (2014) have shown strong dependency of the operating temperature on the TRS yield and reported cellulose decomposition at higher temperature i.e., 230–280 °C with the studies on hydrolysis of sugar cane bagasse in a semi batch type reactor. It has been also reported that, released of sugars from cellulose began at 200 °C, whereas in the present study it was observed at 170 °C. Similar results were also obtained by Lu and Saka (2010) during their study on the hydrolysis of Japanese beech (*F. crenata*) in a batch reactor at 170 °C, whereas in semi batch, temperature was 210 °C. Hence, from this study it can be said that batch reactor allows possible use of lower operating temperatures which can give safe and economically favourable operation. Use of lower operating temperatures would also help in obtaining higher TRS yield.

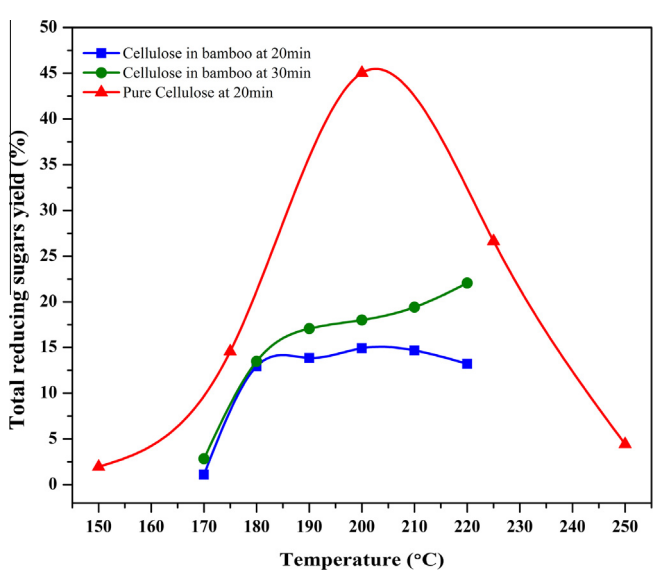

Fig. 4. Comparison of total reducing sugars yield from bamboo (cellulose in bamboo) and pure cellulose at different temperatures.

3.4. Structural changes in bamboo before and after SCW treatment

3.4.1. Fourier transforms infrared (FTIR) spectroscopy

FTIR spectroscopy was used as an analytical tool to investigate the chemical changes in the bamboo structure and to understand the solubilisation process. The FTIR spectra of untreated and treated (180 °C, solid residue) bamboo by SCW treatment is shown in Supplementary Fig. S.4. The FTIR peaks related to cellulose, hemicellulose, and lignin are confirmed by comparing with the available literature data (Steward et al., 1995; Ozturk et al., 2010; Ju et al., 2011). Apparent differences in the transmittance and shapes of the band and their location can be detected in the FTIR spectra. The most representative bands are summarised as follows. The characteristic signals in the FTIR spectrum exhibit strong bonds associated with hydrogen bonded O–H stretching around 3305–3227 cm^{-1} and a major C–H stretching peak absorption at around 2901–2933 cm^{-1}. The characteristics peak at 1736.62 cm^{-1} is assigned to carbonyl (C=O) stretching in hemicellulose structure (Sun et al., 2002). The lignin characteristic peaks at 1590 cm^{-1} and 1512 cm^{-1} represent aromatic C–O stretching vibration; 1447 cm^{-1} represent C–H bending, and peak at 1322 cm^{-1} attributed to C–O absorption (Faix, 1992; Pandey, 1999).

FTIR spectra exhibits C–O absorption peak at 1123 cm^{-1} and 931 cm^{-1} which represents the presence of crystalline and amorphous cellulose in the sample. Appearance of peaks at 1415 cm^{-1}- and 1172 cm^{-1} represents C–H vibrations of cellulose and anti-symmetric stretching C–O–C glycoside bands, respectively (Steward et al., 1995; Liu et al., 2005). The bands at 3435 cm^{-1}, 2893 cm^{-1}, 1184 cm^{-1}, 1113 cm^{-1}, 1065 cm^{-1}, 874 cm^{-1} are associated with cellulose and hemicellulose core structures. The changes in these band intensities indicate that the structure of bamboo was delignified and solubilised after SCW treatment.

3.4.2. X-ray diffraction (XRD)

XRD analysis was performed to estimate the crystallinity index of cellulose in the treated and untreated biomass samples at different temperatures (Supplementary Fig. S.5). The crystallinity index of untreated bamboo was found to be 50.55% and after SCW treatment, significant changes were observed in the crystallinity index. After hydrolysis, crystallinity index of the treated sample was found to be increased from 65.83% to 71.9% with an increase in temperature over the range 170 °C to 210 °C at 40 min hydrolysis time. Crystallinity index value was slightly decreased (70.92%) with further increase in temperature (220 °C, see Fig. 2a). Increase in crystallinity index of treated sample indicates that the hydrolysis effect on amorphous zone is more compared to crystalline zone (Sindhu et al., 2014). This may be attributed to effective removal of certain amount of lignin and hemicellulose from

bamboo structure and not necessarily due to changes in the crystalline structure of the bamboo. The decrease in crystallinity index at 220 °C implies that the treated sample is associated with cellulose decrystallisation, which might be due to phase change took place between 210 and 220 °C (Kumar et al., 2010; Ozturk et al., 2010).

3.4.3. Scanning electron microscopy (SEM)

The SEM analysis of untreated and treated bamboo sample was performed to notice the morphological structural changes occurred during the SCW treatment. SEM images of untreated biomass sample showed highly ordered and rigid fibres (Supplementary Fig. S.6a). The SEM analysis of the sample before SCW treatment shows the presence of carbohydrates, waxes and lignin on the surface and similar pattern was noticed by Ozturk et al. (2010), Ninomiya et al. (2013) during their study on the kenaf (*Hibiscus cannabinus* L.) stems and bamboo powder as a feedstock. Comparing the untreated sample image with the insoluble fraction of treated sample after SCW condition at 180 °C (Supplementary Fig. S.6b), clear cracks can be seen on the surface of hydrolysed biomass at about 1000× magnification. Even the cracks are not steadily dispersed over the sample surface. From the SEM analysis, distorted structure of the treated samples was observed, which improves hydrolysis efficiency.

3.5. Reaction kinetic model for TRS formation and decomposition under sub critical condition

Hydrolysis of the lignocellulosic biomass is a very complex process and results in the formation of various decomposed products. Therefore, a simplified kinetic model is required to estimate its decomposition kinetics. The mechanism of TRS formation and its decomposition can be simplified based on the irreversible series and consecutive reaction as shown in Eq. (2) (Rogalinski et al., 2008; Zhao et al., 2009; Aida et al., 2010; Lin et al., 2015).

$$\text{Biomass (A)} \xrightarrow{k_1} \text{Total reducing sugars (B)} \xrightarrow{k_2} \text{Degraded products (C)} \tag{2}$$

From the above reaction, k_1 (min^{-1}) and k_2 (min^{-1}) are the first-order reaction rate constants of biomass hydrolysis (TRS formation) and degraded products (TRS decomposition). Reaction rate equation for the first step is shown below (Eq. (3)) and after integrating in Eq. (4).

$$r_A = \frac{dC_A}{dt} = -k_1 C_A \tag{3}$$

$$C_A = C_{A0} \exp(-k_1 t) \tag{4}$$

$$r_B = \frac{dC_B}{dt} = k_1 C_A - k_2 C_B \tag{5}$$

$$r_C = \frac{dC_C}{dt} = k_2 C_C \tag{6}$$

$$C_{A0} = C_A + C_B + C_C \tag{7}$$

where C_A is the biomass concentration (mg ml^{-1}), C_{A0} is initial concentration of biomass (mg ml^{-1}), C_B is the total reducing sugars concentration (mg ml^{-1}) and C_C is the degradation product concentration. 't' is the reaction time under subcritical condition (min). The simplified form of Eq. (4) given below.

$$\ln\left(\frac{C_A}{C_{A,0}}\right) = \ln(1 - X_A) = -k_1 t \tag{8}$$

The plot of $-\ln(1 - X_A)$ vs t gives the linear fitting with slope equal to reaction rate constants, k_1 (min^{-1}) of first step (TRS formation).

Reaction rate constant for second series reaction i.e., TRS decomposition (k_2) is demonstrated in Eq. (9).

$$r_B = \frac{dC_B}{dt} = k_1 C_A - k_2 C_B \tag{9}$$

Combining Eqs. (4) and (9) gives 1st order linear differential equation. Taking into account the following boundary conditions at $t = 0$, $C_{B,0} = 0$, Eq. (10) is solved which results in Eq. (11).

$$\frac{dC_B}{dt} + k_2 C_B = k_1 C_{A,0} \exp(-k_1 t) \tag{10}$$

$$C_B = \frac{k_1 C_{A,0}}{k_2 - k_1}(\exp(-k_1 t) - \exp(-k_2 t)) \tag{11}$$

Eq. (11) represents the reducing sugar concentration based on the initial amount of biomass, assuming the series reaction (Eq. (2)). Rate constant k_2, can be obtained from Eq. (11) by using non-linear least square regression analysis by MATLAB. The correlation of reaction rate constants with the temperature can be described by the Arrhenius equation.

$$k = A \exp\left(\frac{-E_a}{RT}\right) \tag{12}$$

where k is the reaction rate constant (min^{-1}) at corresponding temperature, E_a is the activation energy (kJ mol^{-1}), T is thermodynamic temperature of reaction (K), A is the pre-exponential factor (min^{-1}) and $R = 8.314$ J mol^{-1} K^{-1}, Universal gas constant. By simplifying the Eq. (12), Eq. (13) is obtained as:

$$-\ln(k) = \frac{E_a}{RT} - \ln(A) \tag{13}$$

The plot of $-\ln(k)$ vs $1/T$ leads to a straight line with slope of the fitting line indicating the activation energy and pre-exponential factor is determined by the intercept of fitting curve.

3.5.1. Reaction kinetics parameter estimation

The hydrolysis behaviour of treated bamboo was studied over the varying temperature range from 170 °C to 220 °C. The curve fitting for a SCW treatment at 180 °C is shown in Fig. 5. Slope of the straight line indicates the reaction rate constant of TRS formation

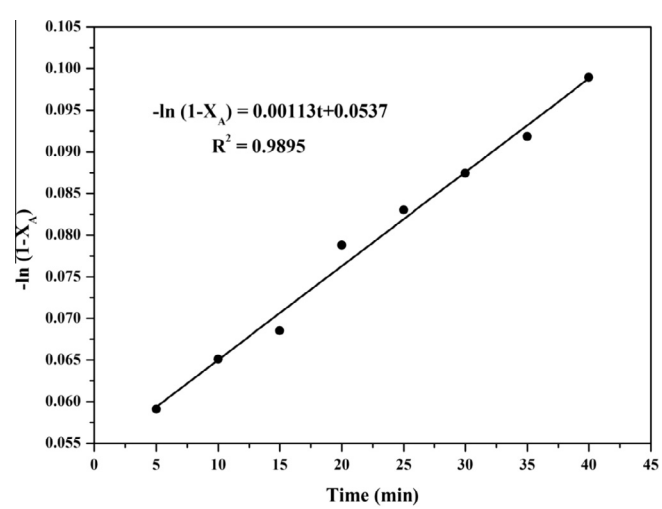

Fig. 5. Fitting curve of reaction rate equation of bamboo hydrolysis at 180 °C and reaction time 40 min.

Table 1
Reaction rate constants of total reducing sugars formation (k_1) and decomposition (k_2) as a function of temperatures.

(a) At reaction time 25 min

Rate constant	170 °C	180 °C	190 °C	200 °C	210 °C	220 °C
k_1 (min^{-1})	1.1×10^{-3}	1.23×10^{-3}	1.5×10^{-3}	1.75×10^{-3}	1.8×10^{-3}	1.9×10^{-3}
k_2 (min^{-1})	9.9×10^{-4}	1.1×10^{-3}	1.3×10^{-3}	1.5×10^{-3}	2.4×10^{-3}	3.5×10^{-3}

(b) At reaction time 40 min

Rate constant	180 °C	190 °C	200 °C	210 °C	220 °C
k_1 (min^{-1})	1.14×10^{-3}	1.33×10^{-3}	1.42×10^{-3}	1.54×10^{-3}	1.74×10^{-3}
k_2 (min^{-1})	1.1×10^{-3}	2.8×10^{-3}	4.5×10^{-3}	6.33×10^{-3}	2.5×10^{-2}

at corresponding temperature and reaction time (40 min). The rate constant for TRS formation at other temperatures can be obtained in the similar way.

In this study, fitting of the kinetic equation was performed in two different ways. In the first case, fitting was carried out considering maximum TRS production and minimum decomposition (degraded products) at an optimum reaction time (i.e., 25 min, 180 °C). While, in the second case, entire reaction time (40 min) (i.e., sugar formation and decomposition) was considered for the model fitting. The rate constants of reducing sugar production and decomposition are estimated from Eqs. (8) and (11).

Table 1a represents the rate constant values for formation of reducing sugar and decomposition products at 25 min reaction time, which was considered as the optimum time to yield maximum reducing sugar at 180 °C. This shows that reaction rate constant values have been positively influenced by the temperature. Similarly, the curve fitting up to total reaction time of 40 min (Table 1b) showed same trend as that of 25 min i.e., with the increase in temperature rate constant values increases (Supplementary Figs. S.7 and S.8). Similar trend was noticed by Rogalinski et al. (2008), Lin et al. (2015) during their study on different kinds of biopolymers and rice straw. Which means that, the hydrolysis rate of bamboo biomass under SCW condition is more at higher temperature compared to a lower temperature; hence chances of formation of degraded products are more at higher temperature.

The plot of $-\ln(k)$ vs $1/T$ was used to estimate the activation energy and pre-exponential factor values of biomass hydrolysis under SCW condition (Supplementary Figs. S.9 and S.10). The rate constants determined at different temperatures can be represented by the straight line. The activation energy and pre-exponential factor for the TRS formation were obtained as 17.97 kJ mol^{-1}, 0.154 min^{-1} at 25 min and 19.25 kJ mol^{-1}, 0.190 min^{-1} at 40 min of reaction time, respectively (Supplementary Table S1). The result of bamboo hydrolysis in the present study recommends that the dissolution of bamboo biomass in SCW condition takes place at comparatively lower temperature (180 °C) in batch type reactor than the reported temperature values in the literature on biomass hydrolysis by SCW treatment (Aida et al., 2010; Cruz et al., 2012). Further, there is a scope to improve the TRS yield by reducing the hydrolysis time and the work is in progress.

4. Conclusions

Hydrolysis of bamboo under SCW treatment resulted in the highest yield of TRS as 42.21% at 180 °C and 25 min hydrolysis time. The biomass crystallinity index was increased from 65.83% to 71.9% by varying temperature from 170 °C to 210 °C; further increased in temperature to 220 °C lowered the crystallinity index (70.92%). From SEM analysis, the distorted structure of the treated samples was observed, which improves hydrolysis efficiency. For TRS formation, the activation energy was found to be 19.25, 17.97 kJ mol^{-1} at 40 min and 25 min, respectively. Although, the hydrolysis of bamboo in water occurs at lower temperatures but requires longer reaction time.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.biortech.2015.05.010.

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